PROJECT - 1 • 2026

AI-Powered Solar Customer Support Platform

A multilingual, omnichannel AI system for faster customer support, grounded answers, personalized interactions and continuously updated solar knowledge.

RAG • ChromaDB • FastAPI • Customer Intelligence • Voice • Messaging

By

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Developer, Hacker, System Arch, ...etc

Executive Vision

One AI brain serving every customer channel with the same trusted company knowledge.

Customer-first

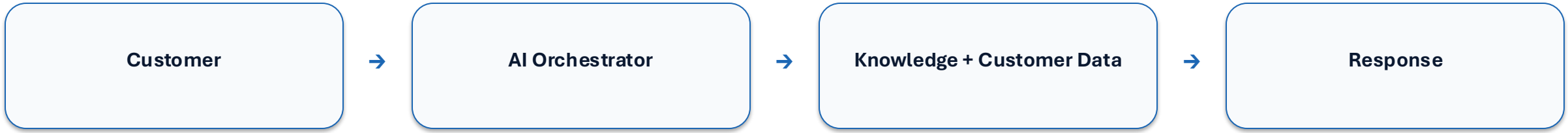
Understand the customer's language, intent, context and emotional state before responding.

Knowledge-grounded

Use ChromaDB RAG to answer from current solar products, manuals, policies and service information.

Omnichannel

Connect web, messaging and phone channels to one reusable AI orchestration layer.



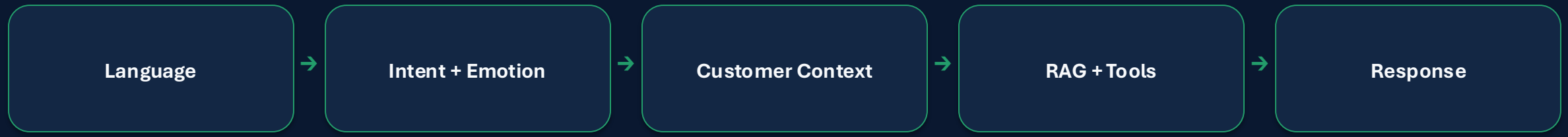
Design principle

Build the intelligence once. Connect every communication channel to the same AI brain.

Solution at a Glance

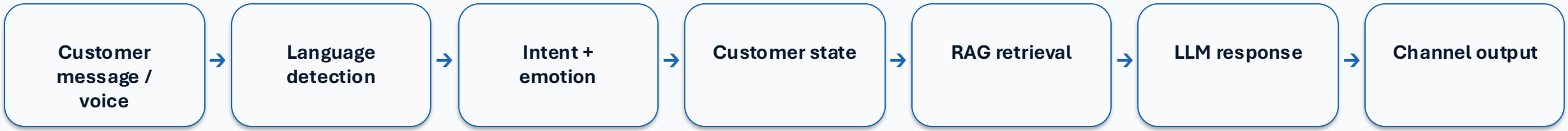
Communication, intelligence, knowledge and business actions in one platform.





End-to-End Customer Interaction

Every channel follows one optimized intelligence pipeline.



Fast path

Common questions can be answered through retrieval, templates or cache without invoking a large model.

Smart path

Complex technical or contextual requests use stronger reasoning with relevant customer and knowledge context.

Human path

Low-confidence, sensitive or high-priority interactions can be routed to a human agent.

Core AI Architecture

A modular architecture keeps the platform fast, maintainable and provider-independent.

CHANNEL LAYER

Web • WhatsApp • SMS • Phone

ORCHESTRATION

Routing • Language • Intent • Emotion • Guardrails

KNOWLEDGE

ChromaDB • Hybrid retrieval • Reranking • Versioning

BUSINESS DATA

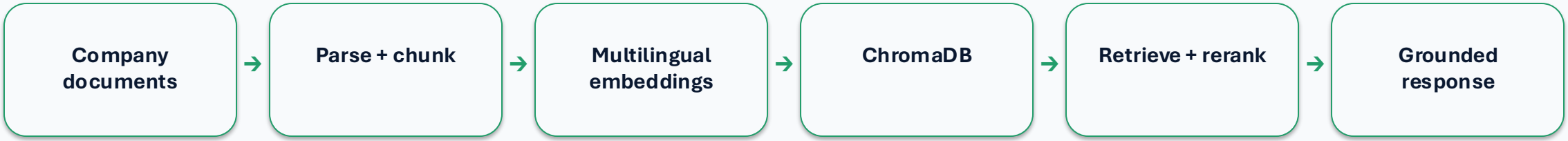
PostgreSQL • Customer • Installation • Tickets • Products

AI SERVICES

Small LLM • Strong LLM • STT • TTS • Embeddings

RAG Knowledge Architecture

ChromaDB is the semantic knowledge layer; structured business state remains structured.



ChromaDB stores

Manuals, FAQs, product specifications, warranty documents, troubleshooting guides and company policies.

PostgreSQL stores

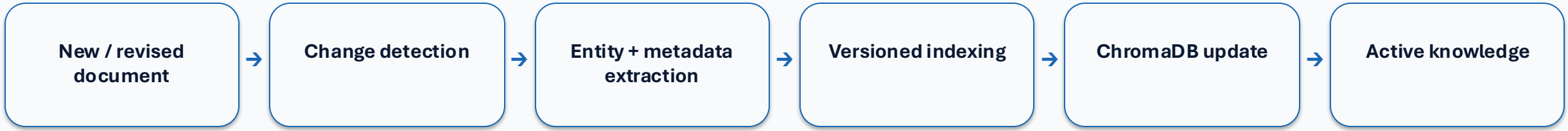
Customer profile, language preference, installation, product assignment, tickets, appointments and operational state.

AI combines both

Relevant company knowledge + current customer state + conversation summary.

Dynamic Knowledge Update Engine

New or changed company information becomes searchable without retraining the LLM.



Example: product discontinued

Product status is changed to DISCONTINUED; active retrieval excludes it from normal recommendations while historical records remain available.

Controlled retrieval

Filter by product, region, effective date, version and active status before the answer reaches the model.

Customer Intelligence & Personalization

Use a small, purposeful customer profile to make every interaction feel continuous.



Short-term memory

Current conversation, active intent and immediate context.

Long-term memory

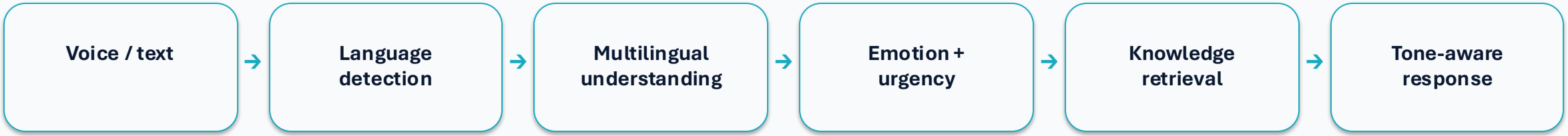
Compact summaries, preferences and service context rather than sending full history.

Business state

Installation, service tickets, appointments and product information from structured data.

Multilingual + Emotional Intelligence

Language and emotional context are first-class signals in the response pipeline.



Languages

Start with English + priority regional languages; expand without redesigning the core architecture.

Emotion

Detect neutral, confused, frustrated, angry, happy and related interaction states.

Urgency

Separate emotional state from urgency so safety/service priority can be handled appropriately.

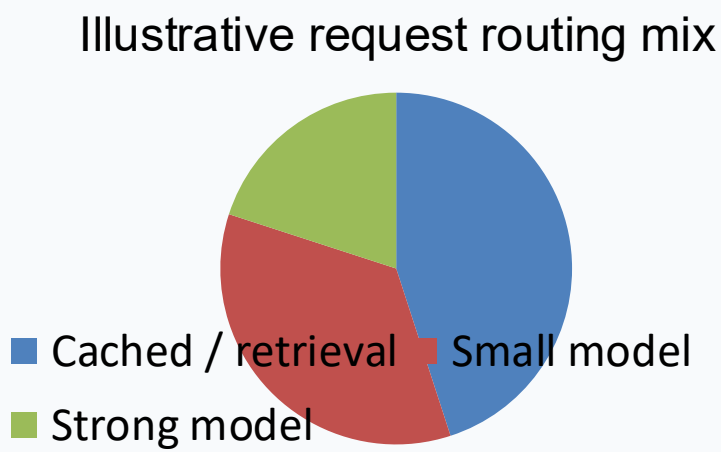
Optimized Model Routing

Spend compute only where intelligence is needed.

Incoming query → Intent / complexity router → Cache / direct answer → Small model → Strong model → Final response

Routing policy

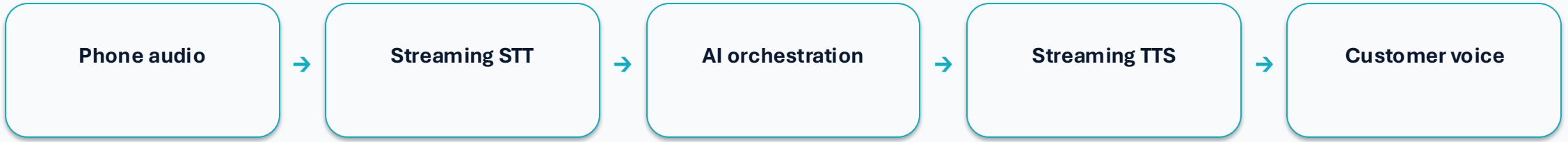
Simple FAQs → cache/retrieval. Normal support → efficient model. Complex troubleshooting → stronger model with RAG and customer context.

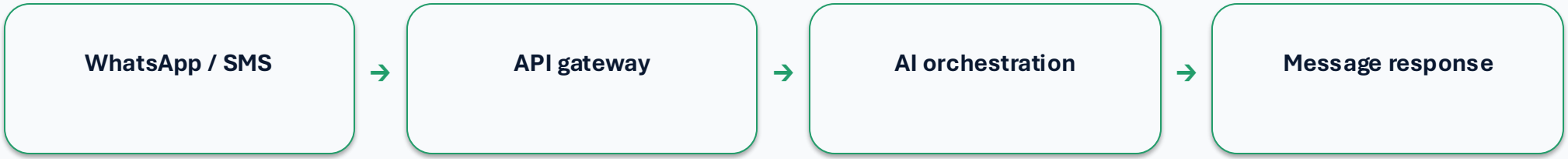


Illustrative planning mix — replace with measured production telemetry after launch.

Voice & Messaging Execution

The same AI brain powers asynchronous messages and real-time voice.





Voice

Streaming speech pipeline for natural turn-taking.(whisper)

Messaging

Persistent context with fast asynchronous responses.

Web

Rich UI, source-aware answers and service actions.

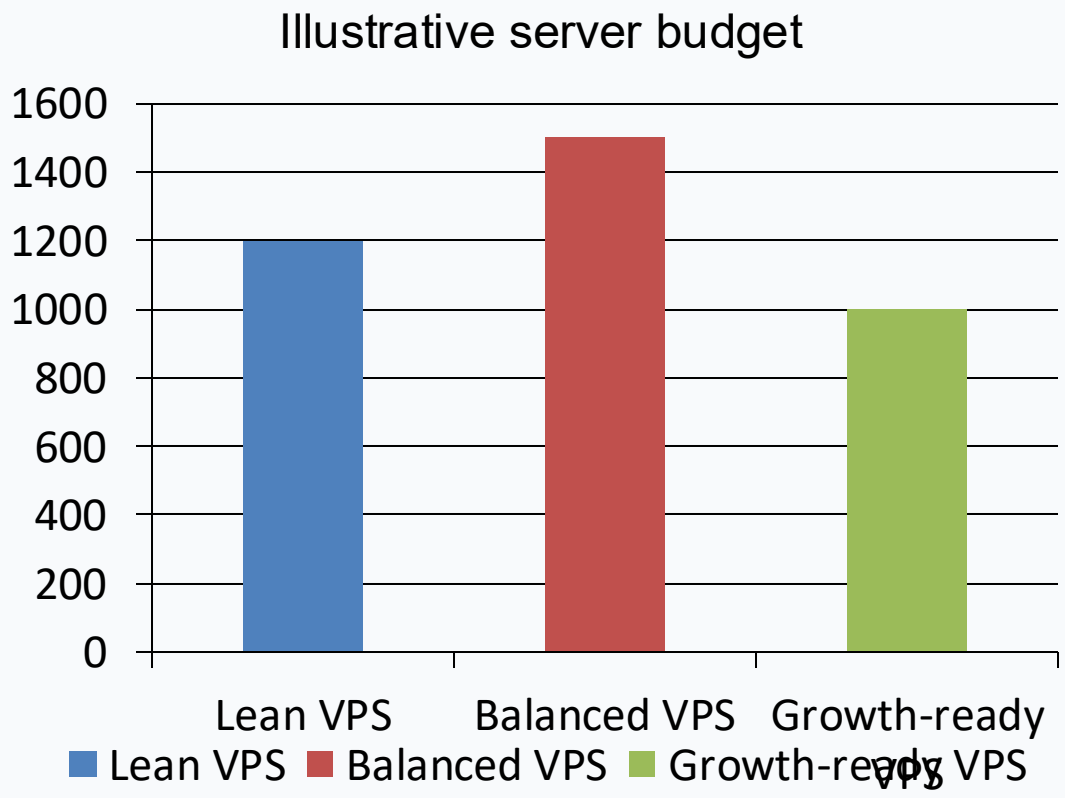
Technology Stack

Prefer open-source components where they reduce recurring cost.

Frontend	React / responsive web	**₹0 software**
Backend	FastAPI + Python	**₹0 software**
Vector layer	ChromaDB	**₹0 self-hosted**
Structured DB	PostgreSQL	**₹0 self-hosted**
Cache / queue	Redis	**₹0 self-hosted**
Embeddings	Multilingual open-source model	**₹0 self-hosted**
AI generation	Efficient LLM + strong fallback	**Usage-based**
Voice	Whisper-family STT + TTS	**₹0 self-hosted / usage-based managed**
Deployment	Cloud + LinuxVPS	**Server cost**
Channels	WhatsApp / SMS / telephony APIs	**Usage-based**

Cost Model — MVP

Indicative planning estimate for a small initial deployment; usage components scale with traffic.



Software

FastAPI, ChromaDB, PostgreSQL, Redis, React and Docker can be self-hosted with ₹0 license cost.

Variable AI

LLM usage depends on message volume, token usage and model routing.

Communication

WhatsApp, SMS and phone costs depend on provider, destination, templates and usage.

Planning target

Small MVP: roughly ₹1,000–₹3,000/month infrastructure + actual usage.

Estimates are planning ranges, not vendor quotations.

Cost Optimization Strategy

Four levers keep recurring AI expenditure under control.

01 • Model routing

Use efficient models for normal traffic and reserve stronger reasoning for complex requests.

02 • Caching

Cache frequent FAQ/retrieval responses and reusable embeddings.

03 • Self-hosted components

Run ChromaDB, PostgreSQL, Redis, embeddings and selected speech services on controlled infrastructure.

04 • Context optimization

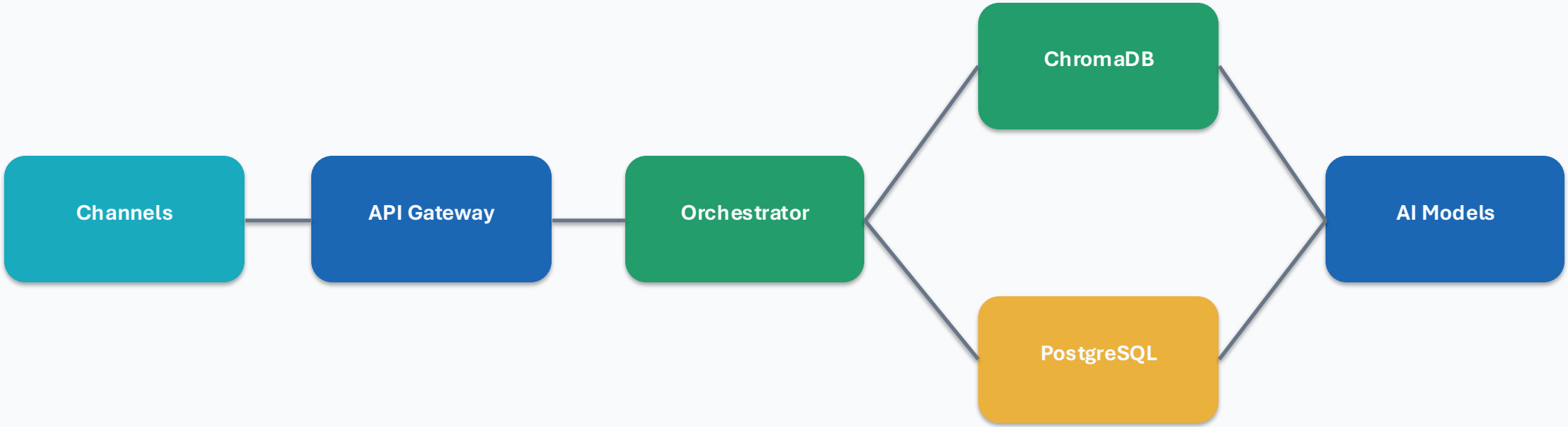
Send summaries and relevant chunks instead of entire conversation histories.

Objective

Minimize cost per resolved customer interaction while preserving answer quality and response speed.

Production Data Flow

A clean separation between knowledge, customer state and AI services enables safe scaling.



One orchestration layer → multiple channels → one consistent customer experience.

Implementation Roadmap

Deliver the platform in controlled, value-focused stages.



Establishing the AI brain first, then connecting channels and business automation.

Operational Intelligence

Continuously measure quality, speed and cost.

Response time

Track p50 / p95 latency

AI cost

Cost per interaction

RAG quality

Retrieval relevance + source coverage

Resolution

AI-resolved interaction rate

Customer experience

Satisfaction + escalation signals

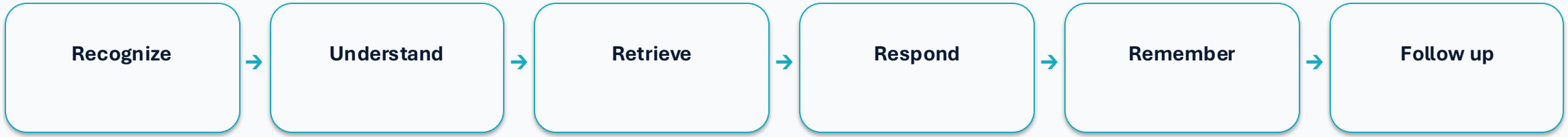
Knowledge freshness

Active document/version coverage

Telemetry closes the loop: measure → tune → deploy → measure.

Target Customer Experience

One knowledgeable, responsive support representative available across channels.



Recognize

Customer identity, language preference, region and current service context.

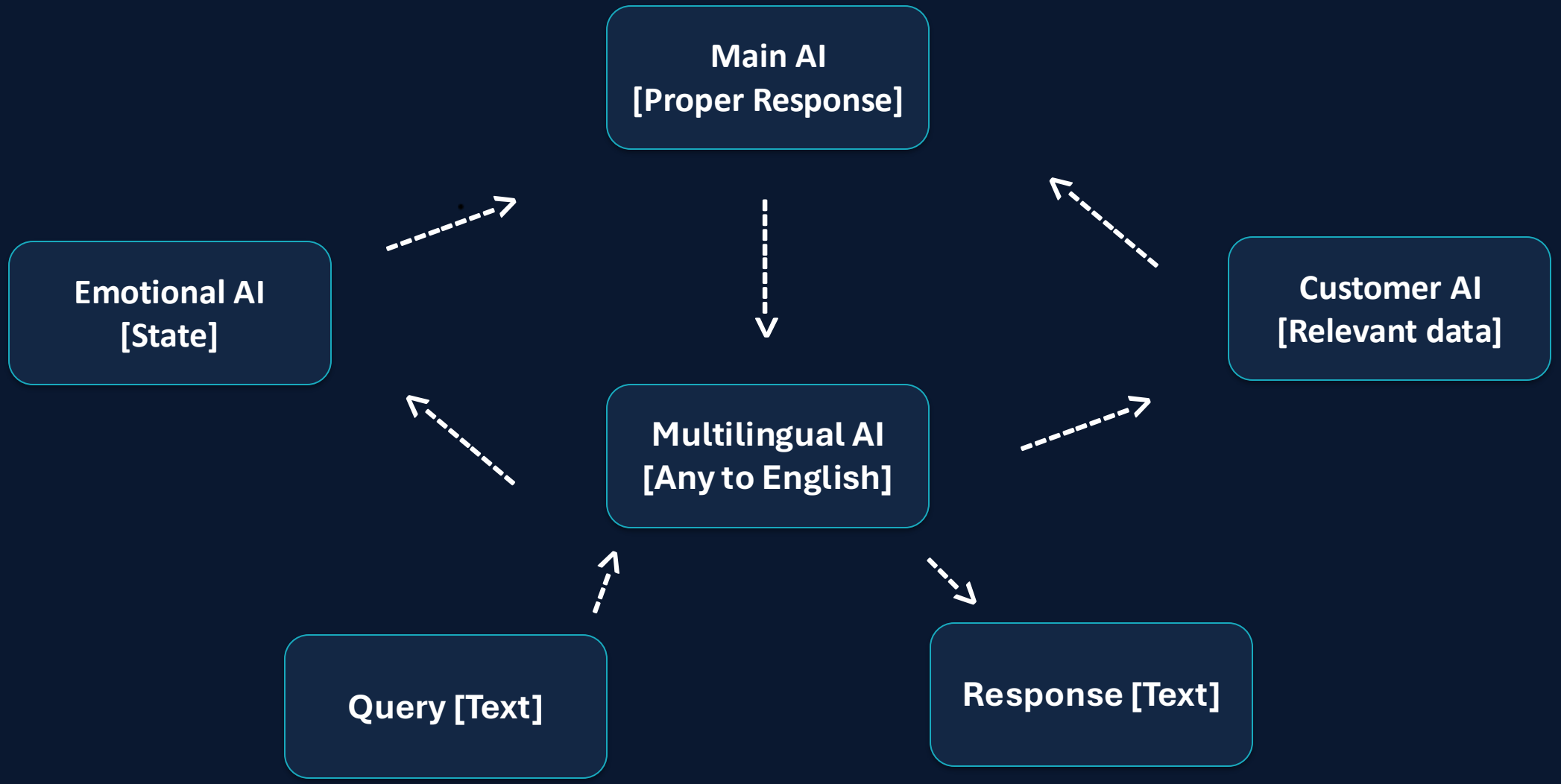
Understand

Intent, emotion, urgency and required business action.

Deliver

Relevant answer, clear next step, consistent tone and appropriate escalation.

AI Model Architecture [open sources]



Open Source Models

Main AI

Qwen3

Emotional AI

Empathetic-qwen3-8b-Jan

Customer AI

Small Qwen 3

MUltiLingual AI

Qwen3.5

STT & TTS

Whisper(optinal) & Qwen3-TTS

Executive Summary

A practical path from AI knowledge base to a complete solar customer-support platform.

AI Brain

One reusable orchestration layer for every channel.

RAG

ChromaDB provides current, searchable company knowledge.

Customer Intelligence

PostgreSQL maintains structured customer and service state.

Optimization

Routing, caching and context compression reduce recurring AI cost.

Omnichannel

Web, messaging and phone connect to the same intelligence.

Continuous Updates

Versioned knowledge ingestion keeps answers aligned with company changes.

Next Step: finalizing the MVP scope, expected traffic and preferred communication providers, then building Phase 1.